Mini-Briefing

Pattern Matching and Machine Learning for Audio Signal Processing MA-INF 2212

#1 Cyclic Operators

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Cyclic Operators: Why?

Example:

„cyclic warp-around“

$$\boxed{T_m x(n) := x(\underbrace{n-m \bmod N}_{\text{„cyclic warp-around"}})}$$

- Easier way of defining & handling operators such as shifts for finite signals
- Compatible with certain math. tools/results, most importantly, the Convolution Theorem for finite signals

  (i.e. the Convolution Theorem for finite signals only works with cyclic convolution)

Time Shifts

1. Infinite signals (non-cyclic case)

Values	...	1	3	4	-5	2	1	…			x
Indexes	…	-1	0	1	2	3	4	5	6	…	
2-shifted values			…	1	3	4	-5	2	1	…	T_2x

$\xrightarrow{2}$

Definition:

$$T_m x(n) := x(n-m)$$

Example:

$$T_2 x(3) = x(3-2) = x(1) = 4$$

Time Shifts

2. Finite signals

Values	[	1	3	4	-5	2	]		
Indexes		0	1	2	3	4		5	6
			2			2			
2-shifted values	[	?	?	1	3	4	]	-5	2

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2. Finite signals

Values	[	1	3	4	-5	2	]			x
Indexes		0	1	2	3	4		5	6	
Cyclic shift	[	-5	2	1	3	4	]	-5	2	T_2x

Definition:

$$T_m x(n) := x(n\text{-}m \bmod N)$$

Example:

$$T_2 x(1) = x(1\text{-}2 \bmod 5) = x(\text{-}1 \bmod 5) = x(4) = 2$$

Time Shifts

2. Finite signals

Values	[	1	3	4	-5	2	]			x
Indexes		0	1	2	3	4		5	6	
Cyclic shift	[	-5	2	1	3	4	]	-5	2	T_2x
Cyclic indexes		5	6	7	8	9				

Time Shifts

$T_m x(n) := x(\,n\text{-}m \bmod N\,)$

2. Finite signals

					…
	-5	-4	-3	-2	-1
Indexes	0	1	2	3	4
Equivalent indexes	5	6	7	8	9
	10	11	12	13	14
	…				

Definition: k mod N := unique $0 \le a < N$ such that $a + rN = k$ for an integer r

Example: 1 = 6 mod 5 = 11 mod 5 = -4 mod 5

Cyclic Autocorrelation

$$\boxed{ACF[x](m) = \langle x | T_m x \rangle}$$

x	=	[	2	0	0	1	3	]		
$T_0 x$	=	[	2	0	0	1	3	]	0	14
$T_1 x$	=	[	3	2	0	0	1	]	1	9
$T_2 x$	=	[	1	3	2	0	0	]	2	2
$T_3 x$	=	[	0	1	3	2	0	]	3	2
$T_4 x$	=	[	0	0	1	3	2	]	4	9

But wait, is cyclicity what we really want?

<u>Cyclic „artifacts“:</u> Also cyclical Patterns are found/detected

$$\boxed{ACF[x](m) = \langle x | T_m x \rangle}$$

x	=	[	2	0	0	1	3	]		m	*ACF[x](m)*
$T_1 x$	=	[	3	2	0	0	1	]		1	9

Compares cyclically shifted „past“
(3) and „present“ (2)

If this is not desirable due to the applications, zero padding of x by *N=length(x)* samples can be used (*x‘ = [2 0 0 1 3 0 0 0 0 0]*)

Cyclic Operators: Conclusions

- Easier way of defining & handling operators such as shifts for finite signals
- Compatible with certain math. tools/results, most importantly, the Convolution Theorem for finite signals

  (i.e. the Convolution Theorem for finite signals only works with cyclic convolution)
- Cyclic shift: samples shifted outside the array to the „right“, are re-inserted to the „left“
- Cyclic Artifacts can be avoided using zero padding